

A Prospective Rank Test of a Detector-Error-Model Weight Screen for Circuit-Level qLDPC Decoding

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1 Motivation and research question

Circuit-level Monte Carlo evaluation is costly across many candidate instances.
We test a DEM-based ranking screen before measuring new logical-block error rates (LERs).

Can DEM structure rank which complete code-circuit-decoder instances BP+OSD decodes more reliably, before new LER results are known?

Connection to classical LDPC decoding

DEM: Tanner graph for syndrome decoding. Logical-block LER: analogue of FER / word-error rate.
Minimum detector-silent logical weight plays a distance-like role. The screen is a verified upper bound.

2 Building the DEM-weight screen

A detector error model (DEM) records which detectors each fault mechanism triggers and which logical observables it flips.

Screen d_{upper} : smallest verified mechanism count found for a pattern that triggers no detectors but flips a logical observable, if the stability gate passes.

Verified upper bound, not code or exact DEM distance.

Before new LER labels

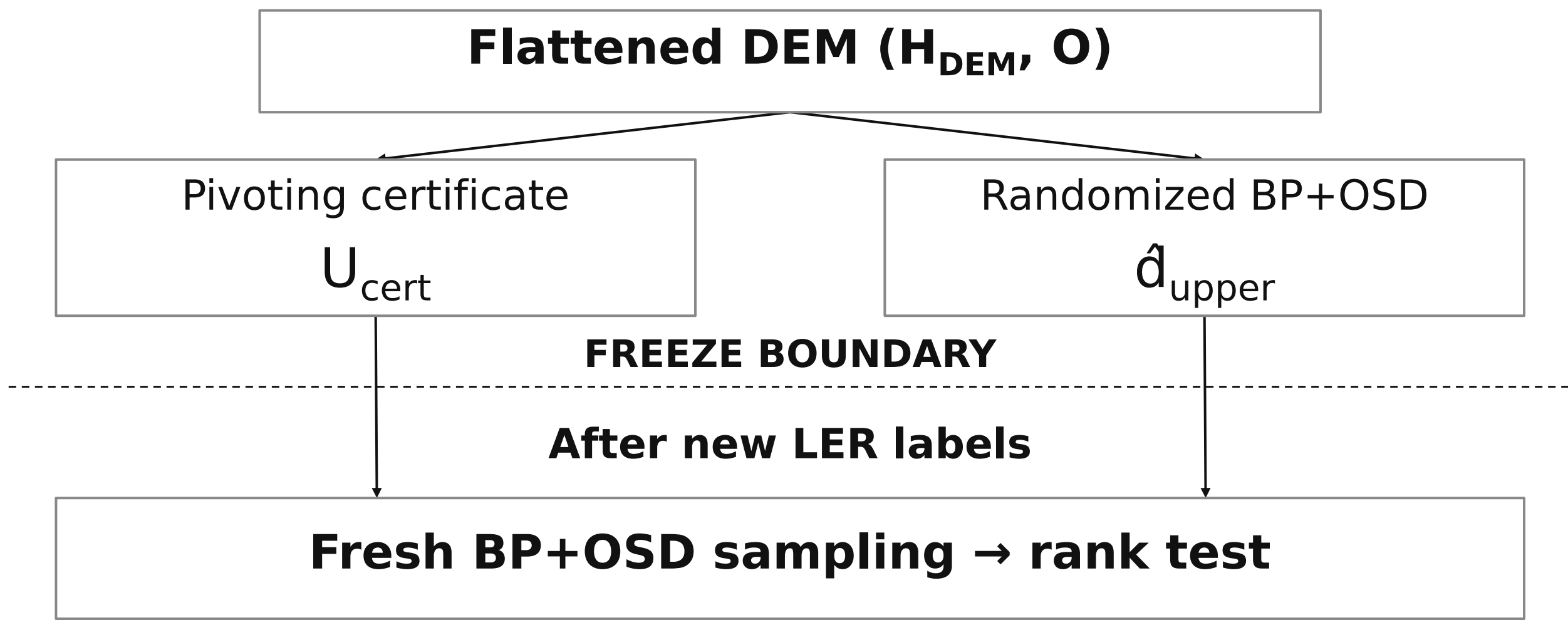


Fig. 1. Structural results freeze before fresh labels.
Two search phases must agree, otherwise abstain.

4 Ranking agreement and evidence

Logical-block LER is the fraction of sampled blocks with any logical observable decoded incorrectly.
Pairs are ordered only when their Wilson 95% LER intervals do not overlap.

Rank hypothesis: higher screen predicts lower LER.

Lower LER is better

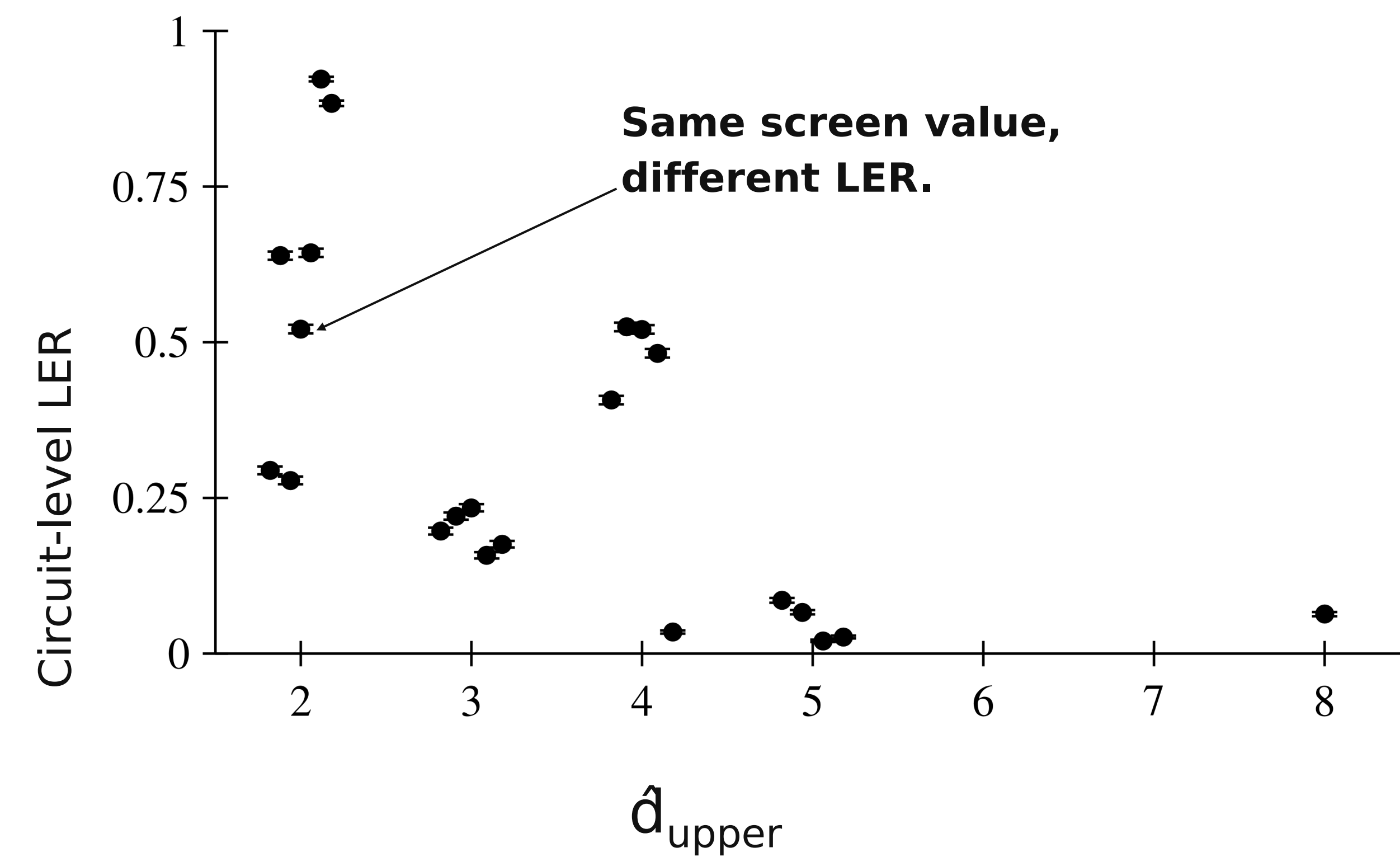


Fig. 2. Wilson 95% bars. Horizontal jitter separates equal integer screen values.

3 The prospective test

Freeze the bank, circuit, decoder, budget and decision rules before new LER labels. Four development anchors are excluded from the confirmatory score.

Table I. Frozen experimental settings (summary)

Item	Frozen value
Population	32 complete instances
Development	BB72, BB90, BB144, BB288
Confirmatory	28 instances; development excluded
Circuit point	p = 0.003; 6 rounds; X-basis memory; SI1000 noise
Schedule	EdgeColoringXZ, smallest-last
Decoder	Product-sum BP, 32 iterations; OSD-CS order 10
Sampling budget	20,000 shots/instance; 640,000 total; no early stop

Table II. Confirmatory pair results (summary)

Quantity	Value
Eligible instances	22
Possible / Wilson-ordered pairs	231 / 226 (97.8%)
Unordered pairs	5
Strict concordance	148 (65.5%)
Screen ties	45 (19.9%)
Discordance	33 (14.6%)

Percentages use the 226 ordered pairs.

Screen coverage and abstentions

Overall: 24/32 ranked, 8/32 abstentions.
Confirmatory: 22/28 ranked (78.6%), 6/28 abstentions.
Abstention = no rank assigned, not a decoding failure.

C = 0.7544 · Ranking agreement with half credit for ties

Positive

$C = (148 + 0.5 \times 45) / 226$. The no-information reference is 0.5.
95% code-resampling stability interval: [0.6262, 0.8382]. Stability to eligible-bank composition.
Positive rule met: ≥ 20 ordered pairs and lower bound > 0.5 . Supports coarse instance ranking at this point.

5 Conclusion and future work

Conclusion

Coarse ranks: 22 instances, $C = 0.75$.
Potential use: candidate triage.
No absolute LER prediction.

Scope

One point, BP+OSD-dependent screen.
Coarse values 2-8 (45 tied pairs).
6/28 confirmatory instances unranked.

Future work

Test new points, decoders and independent instance banks.
Benchmark unproven cost savings.